Design and Topology Optimization of a Compliant Gripper

A2 Design assignment

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1 Introduction

When defining an engineering problem for computational design, aligning functional requirements with physical manufacturability is paramount. Topology optimization has historically been utilized to minimize compliance and weight in rigid structures. However, its application has expanded to the design of compliant mechanisms comprised of pliable polymer materials. These structures achieve mobility through controlled elastic deformation rather than discrete, articulated joints. This approach eliminates backlash, friction, and the need for assembly, making it highly advantageous for precision robotics and Additive Manufacturing (AM) [1].

This project focuses on the computational design and physical realization of a compliant gripper specifically tailored to grasp a hexagonal profile (e.g., a nut or bolt). The primary challenge lies in the non-intuitive nature of compliant design: the algorithm must synthesize a structure that balances sufficient flexibility to actuate with adequate stiffness to transmit force. Furthermore, the theoretically optimal design must be physically viable. Often, topology optimization yields fragile structures that perform well in finite element (FE) simulation but fail during fabrication. By implementing a density-based topology optimization workflow in Python and applying specific algorithmic filters, this project exposes the intricacies of practical topology optimization.

Furthermore, this report will serve as a concise description of the practical implementation of topology optimization. The following sections will: firstly, define the approach to topology optimization; secondly, showcase the approach to realization and manufacturing of the part. This is followed by a brief analysis of the parameters defined for the algorithm, and the performance of the final part. Finally, several takeaways and lessons learned are presented on the usage and implementation of topology optimization algorithms.

2 Methodology

Implicit to the functioning of topology optimization algorithms, is the implementation of a finite element analysis, to model structural compliance to loading cases. This section outlines the finite element subroutine, providing the boundary conditions, constraints, and objective function implemented.

2.1 Finite Element Discretization and Material Interpolation

The compliant mechanism is modeled within a 2D rectangular design space. To maximize computational efficiency, geometric symmetry is exploited along the horizontal mid-plane. The continuous design domain is discretized into a uniform grid utilizing four-node quadrilateral (Q4) finite elements.

The continuous density variables are mapped to the physical stiffness of the elements using the Solid Isotropic Material with Penalization (SIMP) interpolation scheme [2, 3, 4]. This relationship is defined as:

$$E(\rho_e) = E_{min} + \rho_e^p(E_0 - E_{min}) \tag{1}$$

where E_0 is the Young's modulus of the solid base material, E_{min} is a highly diminished stiffness value assigned to void elements to prevent numerical singularities, ρ_e is the physical element density ranging between 0 and 1, and p is the penalization power. A penalization factor of $p = 3.0$ is utilized to disproportionately penalize intermediate "gray" densities [2].

2.2 Boundary Conditions, Constraints, and Objective Function

The mechanism performs a rotational transformation on the actuation load vector: a horizontal compressive input force is applied at the central actuator node, and the objective is to maximize the inward transverse clamping displacement at the jaws. For the sake of computational performance, the modelled grid includes a symmetry condition at the top edge (y=0), removing y-displacement along this edge. The figures created are mirrored around this symmetry, providing the full mechanism. Additionally, a pin constraint was included on the bottom-left corner to provide necessary reaction forces, while allowing rotation. Furthermore, several regions are fixed to either solid or void, demarcating the design space for the algorithm.

The objective function is defined to minimize the negative output displacement, to fit a minimization problem:

$$\min_{\boldsymbol{\rho}} \quad c(\boldsymbol{\rho}) = -\mathbf{l}^T \mathbf{u} \tag{2}$$

An adjoint method is employed here to efficiently compute the sensitivity of the objective function with respect to the design variables $\boldsymbol{\rho}$. Given the equilibrium equation $K\mathbf{u} = \mathbf{f}$, the total derivative of the objective $c = -\mathbf{l}^T\mathbf{u}$ with respect to density ρ_e is derived by introducing an adjoint vector $\boldsymbol{\lambda}$:

$$\frac{dc}{d\rho_e} = -\mathbf{l}^T \frac{d\mathbf{u}}{d\rho_e} = \boldsymbol{\lambda}^T \frac{d\mathbf{f}}{d\rho_e} - \boldsymbol{\lambda}^T \frac{dK}{d\rho_e}\mathbf{u} \tag{3}$$

By solving the adjoint equation $K\boldsymbol{\lambda} = \mathbf{l}$, the sensitivities are calculated without requiring the computationally expensive derivative of the displacement vector $\frac{d\mathbf{u}}{d\rho_e}$ for every element, allowing for rapid convergence in high-resolution meshes.

Subject to the linear elastic equilibrium state $K(\boldsymbol{\rho})\mathbf{u} = \mathbf{f}$ and a strict global volume constraint $V(\boldsymbol{\rho}) \leq V_{max}$, where V_{max} is set to a desired volume fraction (V_{frac}) of the active design domain. To prevent the optimizer from disconnecting the output nodes to achieve infinite theoretical displacement, artificial springs (k_{spring}) are added to the input and output degrees of freedom.

2.3 Parameter test

	k_{spring}	V_{frac}	r_{min}
Centre	1.0	0.35	4
High	2.0	0.4	6
Low	0.5	0.3	2

Table 1: High-low factorial parameter test

The topology optimization algorithm implementation, defines three distinct variables that are used as tuning parameters: k_{spring}); V_{frac}; r_{min}. It should be noted that the arbitrary definition of the design space can influence resulting geometries. However, the grid size and the relation of constrained volumes to its x-coordinate results in roughly 50% of the height being constrained. It was hypothesised that this would leave ample space for the compliant mechanism, without interference of the grid boundary. Given the degree of inputs, it was deemed feasible to perform a factorial parameter test: testing all combinations, with defined highs-and-lows, see Table 1. Additionally, one iteration of mesh converge was attempted, to see whether solutions are stable and well-behaved.

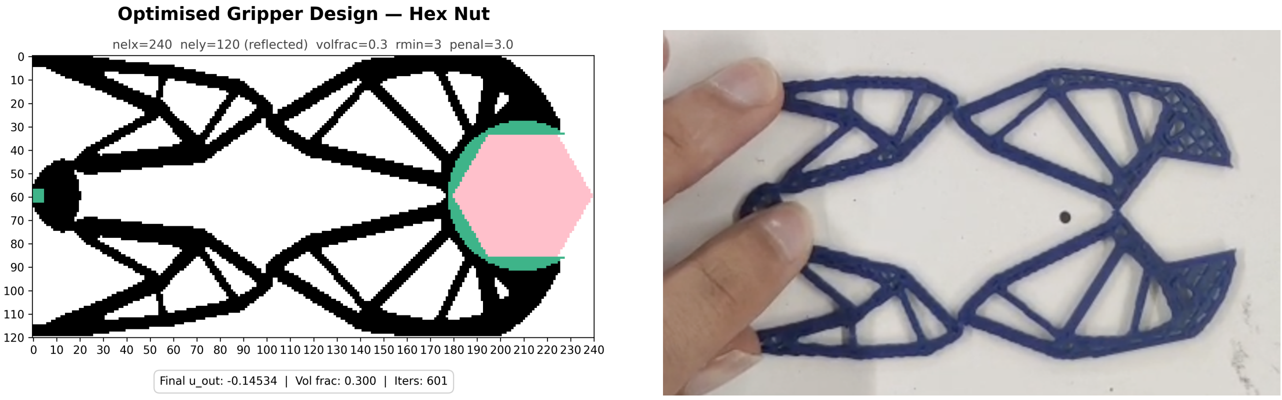
3 Results and Manufacturing Validation

The topology optimization algorithm successfully generated a compliant inverter mechanism. By applying a horizontal input force, diagonal linkage arms are driven forward. Because the outer beams are constrained at the far-left anchors, this forward motion generates a torque that translates into an inward clamping motion at the jaws, effectively gripping the hexagonal profile. The Heaviside projection successfully enforced a distinctly binary structure, allowing for direct extraction to STL format.

The factorial test, showed that resulting designs were greatly sensitive to changes to V_{frac} and r_{min}. The designs were also sensitive to k_{spring}, but no clear feature could be related to this parameter, save for a lack of stability of the algorithm for values outside the magnitude used for the test.

Initially, "Design 3" was selected as printing candidate (because it looked cool). It utilized a volume fraction of 30%, filter radius of $r_{min} = 2.5$ and $k_{spring} = 2$. To test stability and convergence, the mesh resolution was doubled. This resulted in major changes in the produced design, proving that the defined design space contains many attractive local minima. While this design performed well in the finite element simulation, it encountered a failure after production, during removal from the build plate: disconnection at hinge-point. Another simulation was run with r_{min} set to 3, resulting in the design in Figure 1a.

For the sake of a reduction in printing time, it was decided to print the structure only 2mm thick, without any solid layers. Allowing for a short turnover time, provided any errors arise. The final structure, see Figure 1b, could be manipulated in accordance with the FE model[1]



(a) Final optimized topology

(b) Resulting print

Figure 1: Compliant hex-nut gripper design and physical prototype.

4 Conclusion

In this study, a density-based topology optimization framework was successfully implemented to design a compliant hex-nut gripper. By systematically iterating the material distribution within the design domain, the algorithm converged on an optimized layout that balances operational flexibility with structural integrity, achieving the prescribed kinematic behavior through elastic deformation. The transition from the numerical model to a physical prototype underscored critical additive manufacturing constraints. Specifically, enforcing a minimum length scale constraint (r_{min}) proved essential to eliminate checkerboard patterns and ensure that the compliant hinges maintained sufficient thickness for reliable fabrication. The finalized physical print validates the optimization routine, demonstrating compatibility with extrusion-based additive techniques without compromising the mechanism's functional mechanics.

References

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[4] Yin, L., & Ananthasuresh, G. K. *Topology optimization of compliant mechanisms with multiple materials using a peak function material interpolation scheme.* Structural and Multidisciplinary Optimization (2001).

[1] the structure needed some motivation to comply, so there might have been some plastic deformation...oops